Piotr Zalewski I1S6 TI 6.2
Technologie chmurowe Zadanie nr 1 część nieobowiązkowa punkt 2

Linki do repozytoriów z zadaniem:
DockerHub: https://hub.docker.com/r/pzlab33/weather-app/tags
GitHub: https://github.com/pzlab33/zadanie1

Plik kodu aplikacji serwera dla Node.js:

```
const http = require('http');
const https = require('https');

const PORT = 8080;
const AUTHOR = "Piotr Zalewski";

const server = http.createServer((req, res) => {
    const url = new URL(req.url, `http://${req.headers.host}`);

    // Punkt 1b: Prosty UI do wyboru miasta
    if (url.pathname === '/') {
        res.writeHead(200, { 'Content-Type': 'text/html; charset=utf-8' });
        res.end(`
            <html>
                <body style="font-family: sans-serif; text-align: center; padding: 50px;">
                    <h1>Aplikacja Pogodowa</h1>
                    <p>Autor: <b>${AUTHOR}</b></p>
                    <form action="/weather">
                        <select name="city" style="padding: 10px;">
                            <option value="Lublin">Lublin</option>
                            <option value="Warszawa">Warszawa</option>
                            <option value="Londyn">Londyn</option>
                            <option value="Nowy Jork">Nowy Jork</option>
                        </select>
                        <button type="submit" style="padding: 10px;">Sprawdź pogodę</button>
                    </form>
                </body>
            </html>
        `);
    }
    // Punkt 1b: Pobieranie danych pogodowych z zewnętrznego API (wttr.in)
    else if (url.pathname === '/weather') {
        const city = url.searchParams.get('city') || 'Lublin';
        https.get(`https://wttr.in/${city}?format=3`, (apiRes) => {
            let data = '';
            apiRes.on('data', chunk => data += chunk);
            apiRes.on('end', () => {
                res.writeHead(200, { 'Content-Type': 'text/html; charset=utf-8' });
                res.end(`
                    <h2>Pogoda dla: ${city}</h2>
                    <p style="font-size: 24px;">${data}</p>
                    <a href="/">Powrót</a>
                `);
            });
        }).on('error', (err) => {
            res.end("Błąd połączenia z API pogodowym.");
        });
    }
});

// Punkt 1a: Informacja w logach przy starcie
server.listen(PORT, () => {
    const startDate = new Date().toISOString();
    console.log(`[${startDate}] Serwer uruchomiony.`);
    console.log(`Autor: ${AUTHOR}`);
    console.log(`Nasłuchiwanie na porcie TCP: ${PORT}`);
});
```

Plik Dockerfile_zadanie_nr_1:

```dockerfile
# Obraz bazowy z Node.js i Alpine
FROM node:24-alpine

# Metadane zgodnie ze standardem OCI
LABEL org.opencontainers.image.authors="Piotr Zalewski"

# Usunięcie npm (Hardening) - robimy to jako root
RUN rm -rf /usr/local/lib/node_modules/npm \
           /usr/local/bin/npm \
           /usr/local/bin/npx \
           /opt/yarn*

# Kopiowanie kodu aplikacji
COPY --chown=node:node server.js /home/node/server.js

# Dobra praktyka: uruchamianie jako użytkownik bez uprawnień roota
USER node

# HEALTHCHECK (sprawdzenie czy serwer odpowiada)
HEALTHCHECK --interval=30s --timeout=3s --start-period=5s --retries=3 \
  CMD wget --no-verbose --tries=1 --spider http://localhost:8080/ || exit 1

# Dokumentacja portu
EXPOSE 8080

# Uruchomienie aplikacji
CMD ["node", "/home/node/server.js"]
```

Utworzenie buildera opartego na sterowniku docker-container:

```
piotr@tumbleweed:/mnt/Dane 250GB/Studia/Technologie chmurowe/zadanie_nr_1> sudo docker buildx create --name testbuilder --driver=docker-container --use
 --bootstrap
[+] Building 4.7s (1/1) FINISHED
 => [internal] booting buildkit                                                                                    4.7s
 => => pulling image moby/buildkit:buildx-stable-1                                                                 2.9s
 => => creating container buildx_buildkit_testbuilder0                                                             1.8s
testbuilder
piotr@tumbleweed:/mnt/Dane 250GB/Studia/Technologie chmurowe/zadanie_nr_1> sudo docker buildx ls
NAME/NODE          DRIVER/ENDPOINT                   STATUS    BUILDKIT   PLATFORMS
testbuilder*       docker-container
 \_ testbuilder0    \_ unix:///var/run/docker.sock   running   v0.29.0    linux/amd64 (+3), linux/386
default            docker
 \_ default         \_ default                       running   v0.28.0    linux/amd64 (+3), linux/386
piotr@tumbleweed:/mnt/Dane 250GB/Studia/Technologie chmurowe/zadanie_nr_1>
```

Zbudowanie obrazu z cache z użyciem eksportera registry oraz backend-u inline:

```
piotr@tumbleweed:/mnt/Dane 250GB/Studia/Technologie chmurowe/zadanie_nr_1> sudo docker buildx build -f Dockerfile_zadanie_nr_1 -t pzlab33/weather-app:v
1 --platform linux/amd64,linux/arm64 --cache-to type=inline --cache-from type=registry,ref=pzlab33/weather-app:v1 --push .
[sudo] password for root:
[+] Building 43.2s (16/16) FINISHED                                                         docker-container:testbuilder
 => [internal] load build definition from Dockerfile_zadanie_nr_1                                                  0.2s
 => => transferring dockerfile: 841B                                                                               0.0s
 => [linux/arm64 internal] load metadata for docker.io/library/node:24-alpine                                      3.2s
 => [linux/amd64 internal] load metadata for docker.io/library/node:24-alpine                                      3.2s
 => [auth] library/node:pull token for registry-1.docker.io                                                        0.0s
 => [internal] load .dockerignore                                                                                  0.2s
 => => transferring context: 2B                                                                                    0.0s
 => ERROR importing cache manifest from pzlab33/weather-app:v1                                                     1.4s
 => [internal] load build context                                                                                  0.4s
 => => transferring context: 2.32kB                                                                                0.0s
 => [linux/arm64 1/3] FROM docker.io/library/node:24-alpine@sha256:d1b3b4da11eefd5941e7f0b9cf17783fc99d9c6fc34884a665f40a06dbdfc94f  4.9s
 => => resolve docker.io/library/node:24-alpine@sha256:d1b3b4da11eefd5941e7f0b9cf17783fc99d9c6fc34884a665f40a06dbdfc94f  0.1s
 => => sha256:06e652385f83d2c51341a6fc2e3c21185b9e217c062218c20a12a511c8695f55 1.26MB / 1.26MB                     0.4s
 => => sha256:9af8c5455d9150d5ce2e6da2fb0421693187e381ffa1ed5f59b85ef5e9b0f451 444B / 444B                         0.3s
 => => sha256:85317a4b0f43a27c9bda37e67494bf0780e7e46f40a6b7ed8718f7c3e9750c6f 52.39MB / 52.39MB                   2.2s
 => => sha256:d17f077ada118cc762df373ff803592abf2dfa3ddafaa7381e364dd27a88fca7 4.20MB / 4.20MB                     0.7s
 => => extracting sha256:d17f077ada118cc762df373ff803592abf2dfa3ddafaa7381e364dd27a88fca7                          0.2s
 => => extracting sha256:85317a4b0f43a27c9bda37e67494bf0780e7e46f40a6b7ed8718f7c3e9750c6f                          1.6s
 => => extracting sha256:06e652385f83d2c51341a6fc2e3c21185b9e217c062218c20a12a511c8695f55                          0.1s
 => => extracting sha256:9af8c5455d9150d5ce2e6da2fb0421693187e381ffa1ed5f59b85ef5e9b0f451                          0.1s
 => [auth] pzlab33/weather-app:pull token for registry-1.docker.io                                                 0.0s
 => [linux/amd64 1/3] FROM docker.io/library/node:24-alpine@sha256:d1b3b4da11eefd5941e7f0b9cf17783fc99d9c6fc34884a665f40a06dbdfc94f  6.5s
 => => resolve docker.io/library/node:24-alpine@sha256:d1b3b4da11eefd5941e7f0b9cf17783fc99d9c6fc34884a665f40a06dbdfc94f  0.2s
 => => sha256:1ba15d4bb83a5da0244b0ca03df5cac1a19a914c5b876543cb1065d7fddbc9d5 446B / 446B                         0.3s
 => => sha256:3eebacc30b80dca2db4fba20ee2c2562ae69dbbdc307fcee75b49cf859cf63e4 1.26MB / 1.26MB                     0.6s
 => => sha256:d7319775d025f51b0d7ee7f1224231b07f98b38a21909b5a7e2572e083dcb50a 51.51MB / 51.51MB                   2.7s
 => => sha256:6a0ac1617861a677b045b7ff88545213ec31c0ff08763195a70a4a5adda577bb 3.86MB / 3.86MB                     0.7s
 => => extracting sha256:6a0ac1617861a677b045b7ff88545213ec31c0ff08763195a70a4a5adda577bb                          0.2s
 => => extracting sha256:d7319775d025f51b0d7ee7f1224231b07f98b38a21909b5a7e2572e083dcb50a                          1.7s
 => => extracting sha256:3eebacc30b80dca2db4fba20ee2c2562ae69dbbdc307fcee75b49cf859cf63e4                          0.3s
 => => extracting sha256:1ba15d4bb83a5da0244b0ca03df5cac1a19a914c5b876543cb1065d7fddbc9d5                          0.2s
 => [linux/arm64 2/3] RUN rm -rf /usr/local/lib/node_modules/npm            /usr/local/bin/npm            /usr/local/bin/npx            /opt/yar  1.4s
 => [linux/arm64 3/3] COPY --chown=node:node server.js /home/node/server.js                                        0.4s
 => [linux/amd64 2/3] RUN rm -rf /usr/local/lib/node_modules/npm            /usr/local/bin/npm            /usr/local/bin/npx            /opt/yar  0.6s
 => [linux/amd64 3/3] COPY --chown=node:node server.js /home/node/server.js                                        0.3s
 => exporting to image                                                                                            30.0s
 => => exporting layers                                                                                            0.9s
 => => preparing layers for inline cache                                                                           0.1s
 => => exporting manifest sha256:31d7185975052536f705bb9e250e03683b78312cd31a7607eb4d6369075cf821                  0.0s
 => => exporting config sha256:58986f9bab7bb66025a5b5423b8645067988f7f39e0b4415b470a38bf3b9ca98                    0.0s
 => => exporting attestation manifest sha256:12f784470a03e5ebd8b2e2c4fc984921affffa11d0fed6e1c78e0e2257beeae       0.1s
 => => exporting manifest sha256:72e3930fabaf9a04640713f99910a4cac6b4d20b2cc30af8148c19bff767ab3d                  0.0s
 => => exporting config sha256:29f3a9d720cff8b78da69ca579eebd10f7e4a4e1ce6d99220e160c9a4404ae48                    0.0s
 => => exporting attestation manifest sha256:fd639d9d955263468686ca198af78d08cf5b24ec1a958e96ea153bf3d1182062       0.1s
 => => exporting manifest list sha256:a4f256c06308e3b16e8e65debb4f784f150f4fdf6ef3eec5512ae7be238e7415             0.0s
 => => pushing layers                                                                                             23.8s
 => => pushing manifest for docker.io/pzlab33/weather-app:v1@sha256:a4f256c06308e3b16e8e65debb4f784f150f4fdf6ef3eec5512ae7be238e7415  4.7s
 => [auth] pzlab33/weather-app:pull,push token for registry-1.docker.io                                            0.0s
------
 > importing cache manifest from pzlab33/weather-app:v1:
------
piotr@tumbleweed:/mnt/Dane 250GB/Studia/Technologie chmurowe/zadanie_nr_1>
```

Potwierdzenie wykorzystania cache przy ponownym budowaniu obrazu:

```
piotr@tumbleweed:/mnt/Dane 250GB/Studia/Technologie chmurowe/zadanie_nr_1> sudo docker buildx build -f Dockerfile_zadanie_nr_1 -t pzlab33/weather-app:v
1 --platform linux/amd64,linux/arm64 --cache-to type=inline --cache-from type=registry,ref=pzlab33/weather-app:v1 --push .
[sudo] password for root:
[+] Building 12.1s (16/16) FINISHED                                                                                    docker-container:testbuilder
 => [internal] load build definition from Dockerfile_zadanie_nr_1                                                                              0.1s
 => => transferring dockerfile: 841B                                                                                                           0.0s
 => [linux/amd64 internal] load metadata for docker.io/library/node:24-alpine                                                                  1.7s
 => [linux/arm64 internal] load metadata for docker.io/library/node:24-alpine                                                                  1.6s
 => [auth] library/node:pull token for registry-1.docker.io                                                                                    0.0s
 => [internal] load .dockerignore                                                                                                              0.1s
 => => transferring context: 2B                                                                                                                0.0s
 => importing cache manifest from pzlab33/weather-app:v1                                                                                       2.7s
 => => inferred cache manifest type: application/vnd.oci.image.index.v1+json                                                                   0.0s
 => [linux/arm64 1/3] FROM docker.io/library/node:24-alpine@sha256:d1b3b4da11eefd5941e7f0b9cf17783fc99d9c6fc34884a665f40a06dbdfc94f            0.2s
 => => resolve docker.io/library/node:24-alpine@sha256:d1b3b4da11eefd5941e7f0b9cf17783fc99d9c6fc34884a665f40a06dbdfc94f                        0.1s
 => [internal] load build context                                                                                                              0.1s
 => => transferring context: 31B                                                                                                               0.0s
 => [linux/amd64 1/3] FROM docker.io/library/node:24-alpine@sha256:d1b3b4da11eefd5941e7f0b9cf17783fc99d9c6fc34884a665f40a06dbdfc94f            0.1s
 => => resolve docker.io/library/node:24-alpine@sha256:d1b3b4da11eefd5941e7f0b9cf17783fc99d9c6fc34884a665f40a06dbdfc94f                        0.1s
 => [auth] pzlab33/weather-app:pull token for registry-1.docker.io                                                                             0.0s
 => CACHED [linux/arm64 2/3] RUN rm -rf /usr/local/lib/node_modules/npm            /usr/local/bin/npm            /usr/local/bin/npx           /  0.0s
 => CACHED [linux/arm64 3/3] COPY --chown=node:node server.js /home/node/server.js                                                             0.5s
 => CACHED [linux/amd64 2/3] RUN rm -rf /usr/local/lib/node_modules/npm            /usr/local/bin/npm            /usr/local/bin/npx           /  0.0s
 => CACHED [linux/amd64 3/3] COPY --chown=node:node server.js /home/node/server.js                                                             0.5s
 => exporting to image                                                                                                                         6.6s
 => => exporting layers                                                                                                                        0.0s
 => => preparing layers for inline cache                                                                                                       0.3s
 => => exporting manifest sha256:31d7185975052536f705bb9e250e03683b78312cd31a7607eb4d6369075cf821                                              0.0s
 => => exporting config sha256:58986f9bab7bb66025a5b5423b8645067988f7f39e0b4415b470a38bf3b9ca98                                                0.0s
 => => exporting attestation manifest sha256:b3eaf3fc5273f4e1a1c0c9d4523e69b11be79d81ea94d7e9e3332590f5483fc8                                  0.1s
 => => exporting manifest sha256:72e3930fabaf9a04640713f99910a4cac6b4d20b2cc30af8148c19bff767ab3d                                              0.0s
 => => exporting config sha256:29f3a9d720cff8b78da69ca579eebd10f7e4a4e1ce6d99220e160c9a4404ae48                                                0.0s
 => => exporting attestation manifest sha256:0dd0e7ba9f45feda64188dbbbcb95755b78269cc303b6cc1e4887475b3716ed3                                  0.1s
 => => exporting manifest list sha256:064b8526e3e5d9b424cd50c4885de9abce446f9983dc49356ba2d355bd717c88                                         0.0s
 => => pushing layers                                                                                                                          2.2s
 => => pushing manifest for docker.io/pzlab33/weather-app:v1@sha256:064b8526e3e5d9b424cd50c4885de9abce446f9983dc49356ba2d355bd717c88           3.7s
 => [auth] pzlab33/weather-app:pull,push token for registry-1.docker.io                                                                        0.0s
piotr@tumbleweed:/mnt/Dane 250GB/Studia/Technologie chmurowe/zadanie_nr_1>
```

Potwierdzenie zbudowania obrazu na dwie wymagane w zadaniu platformy sprzętowe:

```
piotr@tumbleweed:/mnt/Dane 250GB/Studia/Technologie chmurowe/zadanie_nr_1> docker buildx imagetools inspect pzlab33/weather-app:v1
Name:      docker.io/pzlab33/weather-app:v1
MediaType: application/vnd.oci.image.index.v1+json
Digest:    sha256:064b8526e3e5d9b424cd50c4885de9abce446f9983dc49356ba2d355bd717c88

Manifests:
  Name:        docker.io/pzlab33/weather-app:v1@sha256:31d7185975052536f705bb9e250e03683b78312cd31a7607eb4d6369075cf821
  MediaType:   application/vnd.oci.image.manifest.v1+json
  Platform:    linux/amd64

  Name:        docker.io/pzlab33/weather-app:v1@sha256:72e3930fabaf9a04640713f99910a4cac6b4d20b2cc30af8148c19bff767ab3d
  MediaType:   application/vnd.oci.image.manifest.v1+json
  Platform:    linux/arm64

  Name:        docker.io/pzlab33/weather-app:v1@sha256:b3eaf3fc5273f4e1a1c0c9d4523e69b11be79d81ea94d7e9e3332590f5483fc8
  MediaType:   application/vnd.oci.image.manifest.v1+json
  Platform:    unknown/unknown
  Annotations:
    vnd.docker.reference.digest: sha256:31d7185975052536f705bb9e250e03683b78312cd31a7607eb4d6369075cf821
    vnd.docker.reference.type:   attestation-manifest

  Name:        docker.io/pzlab33/weather-app:v1@sha256:0dd0e7ba9f45feda64188dbbbcb95755b78269cc303b6cc1e4887475b3716ed3
  MediaType:   application/vnd.oci.image.manifest.v1+json
  Platform:    unknown/unknown
  Annotations:
    vnd.docker.reference.digest: sha256:72e3930fabaf9a04640713f99910a4cac6b4d20b2cc30af8148c19bff767ab3d
    vnd.docker.reference.type:   attestation-manifest
piotr@tumbleweed:/mnt/Dane 250GB/Studia/Technologie chmurowe/zadanie_nr_1>
```

Analiza CVE z wykorzystaniem narzędzia Trivy:

```
piotr@tumbleweed:/mnt/Dane 250GB/Studia/Technologie chmurowe/zadanie_nr_1> trivy image pzlab33/weather-app:v1
2026-05-11T00:09:43+02:00       INFO    [vuln] Vulnerability scanning is enabled
2026-05-11T00:09:43+02:00       INFO    [secret] Secret scanning is enabled
2026-05-11T00:09:43+02:00       INFO    [secret] If your scanning is slow, please try '--scanners vuln' to disable secret scanning
2026-05-11T00:09:43+02:00       INFO    [secret] Please see https://trivy.dev/docs/v0.70/guide/scanner/secret#recommendation for faster secret detectio
n
2026-05-11T00:09:45+02:00       INFO    Detected OS     family="alpine" version="3.23.4"
2026-05-11T00:09:45+02:00       INFO    [alpine] Detecting vulnerabilities...   os_version="3.23" repository="3.23" pkg_num=18
2026-05-11T00:09:45+02:00       INFO    Number of language-specific files       num=1
2026-05-11T00:09:45+02:00       INFO    [node-pkg] Detecting vulnerabilities...

Report Summary
```

Target	Type	Vulnerabilities	Secrets
pzlab33/weather-app:v1 (alpine 3.23.4)	alpine	0	-
usr/local/lib/node_modules/corepack/package.json	node-pkg	0	-

```
Legend:
- '-': Not scanned
- '0': Clean (no security findings detected)

piotr@tumbleweed:/mnt/Dane 250GB/Studia/Technologie chmurowe/zadanie_nr_1>
```